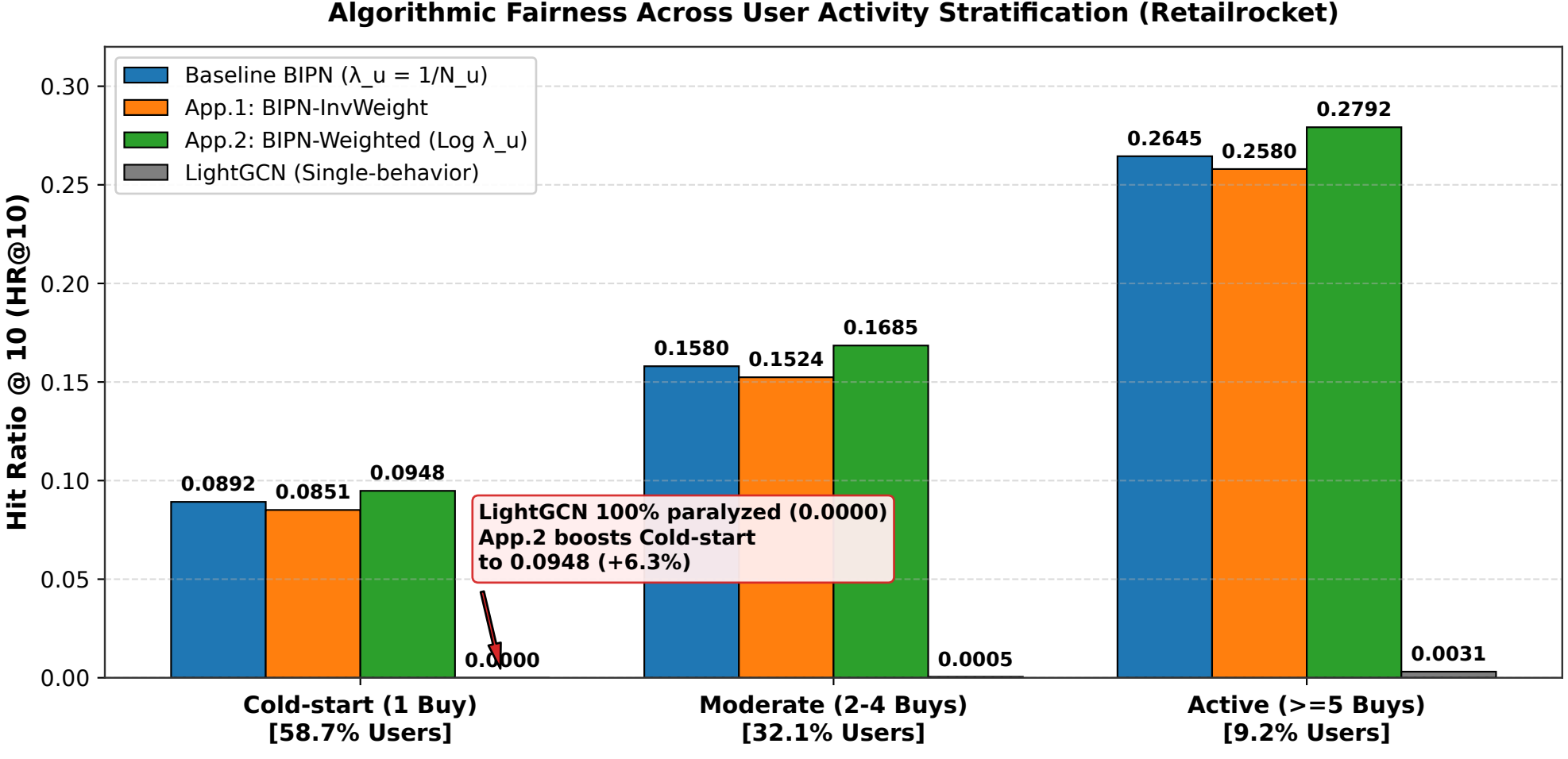
Algorithmic Fairness Across User Activity Stratification (Retailrocket)
Baseline BIPN (λ_u = 1/N_u)
App.1: BIPN-InvWeight
App.2: BIPN-Weighted (Log λ_u)
LightGCN (Single-behavior)
Hit Ratio @ 10 (HR@10)
0.30
0.25
0.20
0.15
0.10
0.05
0.00
0.0892
0.0851
0.0948
0.0000
0.1580
0.1524
0.1685
0.0005
0.2645
0.2580
0.2792
0.0031
LightGCN 100% paralyzed (0.0000)
App.2 boosts Cold-start
to 0.0948 (+6.3%)
Cold-start (1 Buy)
[58.7% Users]
Moderate (2-4 Buys)
[32.1% Users]
Active (>=5 Buys)
[9.2% Users]